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PITCH DECK

ESTIMATING 1-YEAR PD FOR BANCA MASSICCIA

TEAM TURQUOISE

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11/20/2025

BUSINESS UNDERSTANDING

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CREDIT RISK

The potential for financial loss that a lender or creditor faces if a borrower fails to repay a debt or meet their contractual obligations as agreed.

It leads to the following problems:

Reduced profitability: Defaults lead to a loss of interest income and require banks to set aside more money for loan losses, directly decreasing profits.

Increased collection costs: Banks may have to spend more on collection efforts or hire third-party agencies to pursue payment, which can be expensive.

Damaged reputation: A high rate of defaults can damage a bank's reputation, leading to concerns from customers, creditors, and investors.

Due to the above reasons predicting defaults becomes crucial for banks to ensure smooth operations.

Banks need Probability of Default (PD) models to assess and manage credit risk, which is essential for making informed decisions on loan approvals.

STRUCTURAL MODELS

Active Credit Portfolio Management in Practice provides background on several structural credit-risk models that form the foundation for understanding how probability of default (PD) can be modeled.

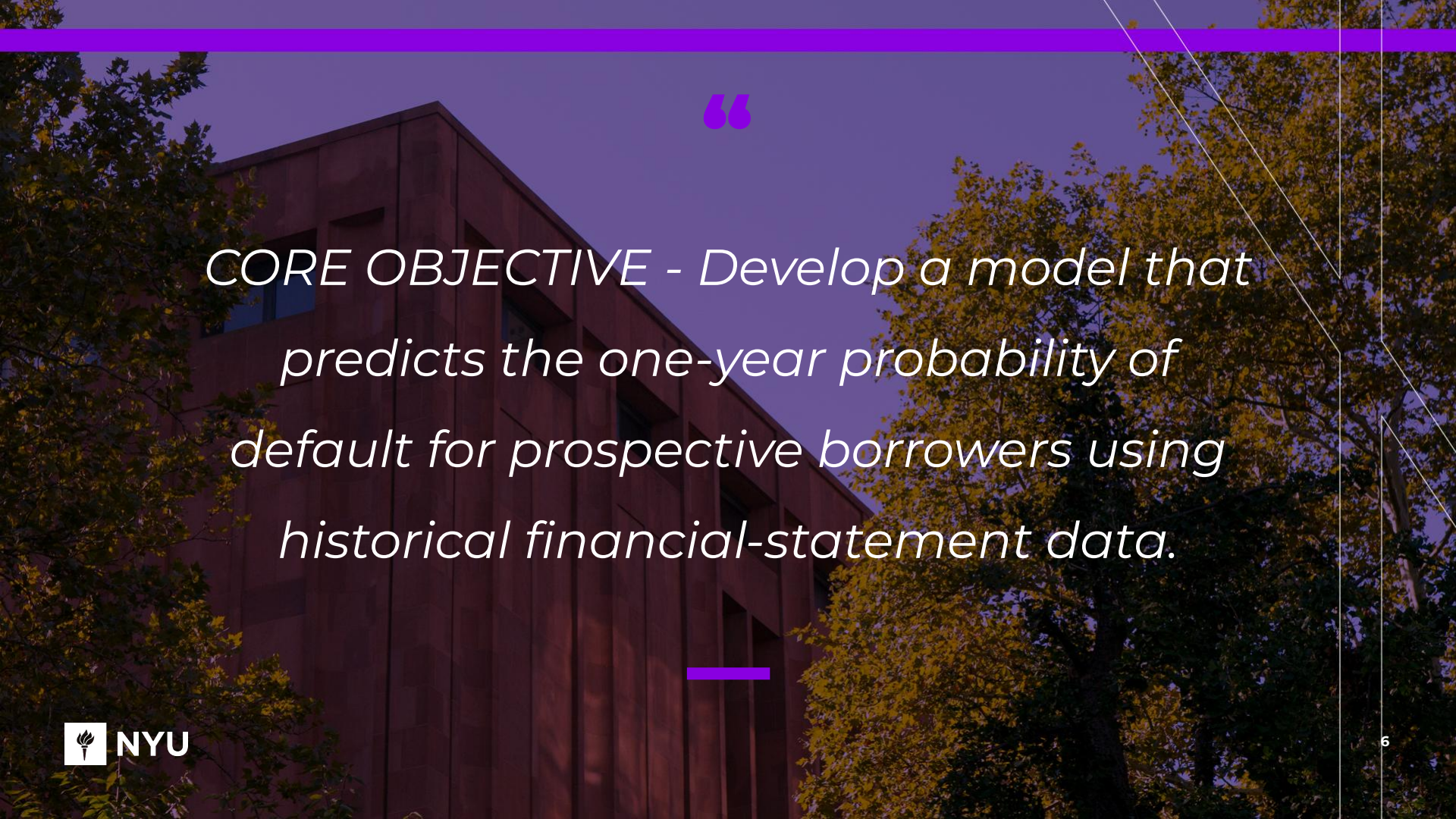
- **Merton / Black-Scholes-Merton (BSM) Model**: This model treats a firm's equity as a call option on its assets. Default occurs only at the debt's maturity date—specifically when the firm's asset value falls below its liabilities.
- **Black–Cox (BC) Model**: Extends Merton by allowing early default. It introduces a default barrier, and if the firm's asset value crosses this barrier at any time before maturity, the firm is considered to have defaulted.
- **Vasicek–Kealhofer (VK / KMV) Model**: Estimates the market value and volatility of a firm's assets. Then computes the Distance to Default (DD). Historical data are used to translate DD into an empirical probability of default (PD).

These structural models rely on equity prices and other market-based valuations. Because our dataset does not include any market information, we shift to using econometric models instead.

RiskCalc APPROACH

The RiskCalc framework, described in *Active Credit Portfolio Management in Practice (Chapter 4: Econometric Models)*, is designed to estimate the probability of default (PD) for private firms. Private-firm financial data are often noisy, incomplete, and exhibit nonlinear relationships with default risk. The RiskCalc methodology addresses these issues using the following steps:

- **Nonparametric Transformation**: Each raw financial ratio is converted into a smooth density-based transformation. This captures the real empirical relationship between the ratio and default risk, even when the raw data are noisy or nonlinear.
- **Logit PD Estimation**: The transformed variables are fed into a logistic regression model, which estimates the probability of default using these stabilized, monotonic predictors.
- **Calibration**: The predicted PDs are then calibrated using a nonparametric mapping so that model estimates match observed default rates. Newer versions also adjust for industry effects.
- **Validation**: The model is tested for out-of-sample accuracy, calibration quality, parameter stability, robustness, and performance metrics such as ROC/AUC.



CORE OBJECTIVE - Develop a model that predicts the one-year probability of default for prospective borrowers using historical financial-statement data.

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DATA UNDERSTANDING

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DATA

Overview of the dataset

- Each row is one firm-year.
- Includes balance sheet, income statement, and financial structure variables.
- All quantities (except ratios) are reported n €.
- Only firms with > €1.5MM in assets are included in data.
- There is no information in the training data regarding defaults that occurred after 12/31/2012.

STRENGTHS

Large historical dataset with many firm-year observations, giving strong statistical power and reliable pattern discovery.

Rich financial statement variables, enabling construction of meaningful ratios (liquidity, leverage, profitability, efficiency) essential for default prediction.

FLAWS

Missing and inconsistent financial data, requiring cleaning, imputation, and smoothing before modelling.

High multicollinearity among raw ratios, leading to redundant predictors that had to be removed using correlation and VIF analysis.

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DATA PREPARATION

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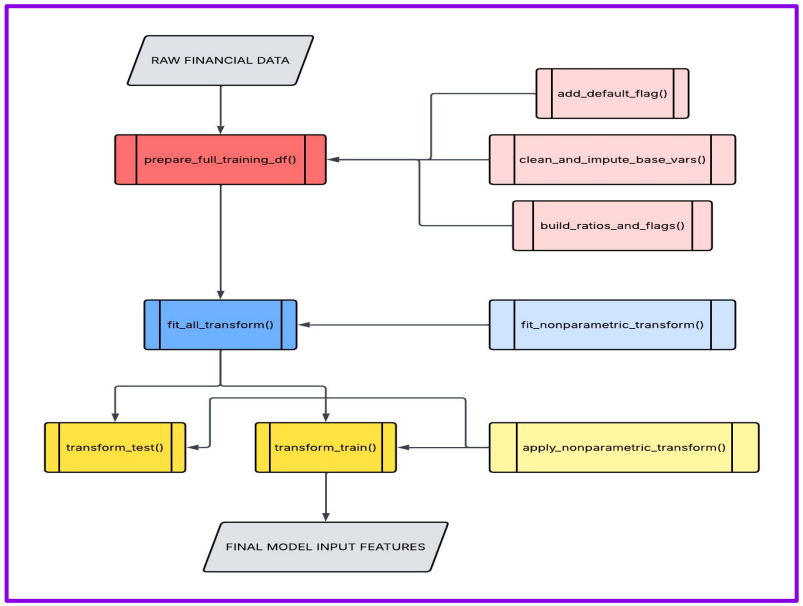
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OVERVIEW

The preprocessing pipeline consists of various functions that can be implemented in a structured and selective way, making the process reproducible.

The workflow consists of:

- Filtering data by year (fs_year)
- Creating target variable (default_flag)
- Cleaning base accounting variables
- Constructing financial ratios and missing flags
- Non-parametric transformations
- Integrating features into a model-ready data-set



A flowchart illustrating all the functions included in the Data Preparation Pipeline.

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TARGET VARIABLE CONSTRUCTION

A **key issue** in default modelling is that financial statements are often reported several months after the fiscal year-end. If a model uses the fiscal year-end date as the information date, it unintentionally **“peeks into the future”**[referenced from *Active Credit Portfolio Management in Practice*].

To address this, we introduce **a reporting delay** so that the model uses only the data that would have been available at that time.

The `add_default_flag()` function creates a binary target variable in a way that is both time-consistent and realistic.

It includes the following steps:

- Apply a 4-month reporting lag (statements available by end-April)
- Define a 12-month prediction horizon
- A firm defaults (1) if: default date ≥ info_date && default date < info_date + 12 months
- Else → 0 (non-default)

	stmt_date	info_date	horizon_end	def_date	default_flag
998976	2009-12-31	2010-04-30	2011-04-30	2013-05-13	0
1010009	2011-12-31	2012-04-30	2013-04-30	2013-04-02	1
979570	2008-12-31	2009-04-30	2010-04-30	2011-12-10	0
972460	2010-12-31	2011-04-30	2012-04-30	2011-10-25	1
964938	2007-12-31	2008-04-30	2009-04-30	2010-11-08	0
998499	2008-12-31	2009-04-30	2010-04-30	2013-05-22	0
988027	2009-12-31	2010-04-30	2011-04-30	2012-03-19	0
978178	2008-12-31	2009-04-30	2010-04-30	2011-09-09	0

Default Flag Creation.

DATA FILTERING AND BASE VARIABLE PREPARATION

Filtering Training Data

The final model is trained on historical financial data available up to **2011**.

The decision is motivated by the following considerations:

- The final evaluation is conducted on 2012 data
- Default events occur only up to May 2012 and subsequent months contain no defaults
- Our 4 month lag would only capture defaults in May (as the info_date shifts to April 2012)

Cleaning Base Variable

1	Redundant Columns	For example, *eqity_corp_family_tot* was entirely empty and therefore removed.
2	Identifying Key Variables	Such as *COGS*, *rev_operating*, *prof_operations*, *cash_and_equiv*, and *roe*, along with assessing their missing values.
3	Targeted Imputation	Missing values for key variables were imputed only when the underlying components required to compute them were available.

CONSTRUCTING FINANCIAL RATIOS

We **identified all ratios** that could be computed from the available data and cleaned the underlying base variables as thoroughly as possible.

Once the base variables were prepared, we organized the resulting ratios into **five** broad factor groups.

Each factor captures a different dimension of a **firm's financial health**.

build_ratios_and_flags() is a wrapper functions which calculates all the ratios and their missing flags.

LEVERAGE	`df["debt_to_equity_raw"] = df["liab_tot"] / df["eqty_tot"]`
	`df["debt_ratio_raw"] = df["liab_tot"] / df["asst_tot"]`
	`df["equity_ratio_raw"] = df["eqty_tot"] / df["asst_tot"]`

PROFITABILITY	`df["operating_margin_raw"] = df["prof_operations"] /`
	`df["rev_operating"]`
	`df["roa_raw"] = df["prof_operations"] / df["asst_tot"]`

EFFICIENCY	`df['asst_turnover_ratio_raw'] = df['rev_operating'] / df['asst_tot']`
LIQUIDITY	`df["current_ratio_raw"] = df["asst_current"] / df["debt_st"]` `df["cash_ratio_raw"] = df["cash_and_equiv"] / df["debt_st"]`

COVERAGE	`df["asst_coverage_ratio_raw"] = (`
	`(df["asst_tot"] - df["asst_intang_fixed"] - df["debt_st"]) /`
	`(df["debt_st"] + df["debt_lt"]))`

SIZE	`df['log_asst_tot'] = np.log1p(df['asst_tot'].clip(lower=0))`

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CONSTRUCTING FINANCIAL RATIOS

Missing-value flags were created for cases where ratio **calculations were not possible**, usually because of zero denominators or missing underlying variables.

These instances were quite uncommon relative to the full dataset.

The resulting raw ratios are **naturally noisy and unstable**, but retaining a missingness flag helps capture any predictive signal associated with the absence of data.

```python
def prepare_full_training_df(df: pd.DataFrame) -> pd.DataFrame:
    # Convenience wrapper for training: add default_flag + clean + ratios.
    # print_header("Preparing full training dataframe (default flag + cleaning + ratios)...")
    df = add_default_flag(df)
    df = clean_and_impute_base_vars(df)
    df = build_ratios_and_flags(df)
    return df
```

Wrapper function which prepares raw training data.

```
Missing Values:
debt_to_equity_raw    1
debt_ratio_raw        1
equity_ratio_raw      1
dtype: int64


Missing Values:
gross_margin_raw          620
operating_margin_raw      620
roa_raw                    21
roe_raw                 38649
dtype: int64


Missing Values:
asst_turnover_ratio_raw    21
dtype: int64


Missing Values:
current_ratio_raw      27
cash_ratio_raw        122
dtype: int64


Missing Values:
asst_coverage_ratio_raw    278
dtype: int64
```

```
Raw ratios:
 - roe_raw
 - debt_to_equity_raw
 - debt_ratio_raw
 - equity_ratio_raw
 - gross_margin_raw
 - operating_margin_raw
 - roa_raw
 - asst_turnover_ratio_raw
 - current_ratio_raw
 - cash_ratio_raw
 - asst_coverage_ratio_raw

Missing flags:
 - roe_raw_missing
 - debt_to_equity_raw_missing
 - debt_ratio_raw_missing
 - equity_ratio_raw_missing
 - gross_margin_raw_missing
 - operating_margin_raw_missing
 - roa_raw_missing
 - asst_turnover_ratio_raw_missing
 - current_ratio_raw_missing
 - cash_ratio_raw_missing
 - asst_coverage_ratio_raw_missing
```

Missing values in each raw ratios and the list of final raw ratios.

FEATURE SELECTION

To ensure the model uses **stable features**, we performed a structured feature-selection process before applying nonparametric transformations.

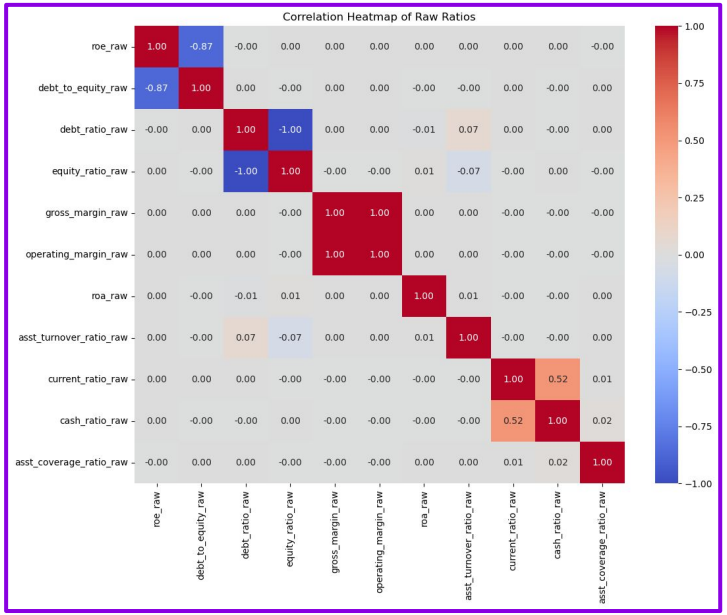
We performed **correlation analysis** and **VIF diagnostics** to identify multicollinearity among financial ratios.

The heatmap showed several ratios with perfect or **near-perfect correlations**, and the VIF table confirmed extreme multicollinearity for variables like *gross margin*, *operating margin*, *equity ratio*, and *debt-to-equity ratio*.

To prevent instability in the logistic regression model, we **removed these redundant features** and their flags.

The final feature set retained only ratios that provide unique predictive information.

FEATURE SELECTION



Correlation heatmap of raw financial ratios showing several pairs with very high correlation, indicating redundant features that may cause multicollinearity.

	feature	VIF
0	roa_raw	1.000067e+00
1	asst_coverage_ratio_raw	1.003700e+00
2	asst_turnover_ratio_raw	1.008649e+00
3	equity_ratio_raw	1.235880e+00
4	current_ratio_raw	1.394988e+00
5	cash_ratio_raw	1.399238e+00
6	debt_ratio_raw	1.465635e+00
7	roe_raw	4.297204e+00
8	debt_to_equity_raw	4.297698e+00
9	gross_margin_raw	3.602880e+14
10	operating_margin_raw	3.602880e+14

VIF scores for raw ratios, highlighting variables with extremely high multicollinearity, supporting their removal from the final feature set.

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NON-PARAMETRIC TRANSFORMATION

$$q^* = q_i + \frac{x - b_i}{b_{i+1} - b_i} \frac{1}{k}$$

In this example, the transformed value of liquidity for a firm with liquidity of 0.015 would fall between the zero bucket and the first bucket. If there were 100 buckets, the probability of default associated with liquidity of 0.015 would be found by first computing

$$q^* = 0.0 + \frac{0.015 - 0.001}{0.02 - 0.001} \frac{1}{100} = 0.00737$$

An example of Non-parametric transformations as discussed in ***Chapter 4*** of ***Active Credit Portfolio Management in Practice***. The value 0.015 lies between bucket bounds 0.001 and 0.02, so its transformed quantile q*q^*q* is computed by linear interpolation using the formula shown.

Need for Non-parametric transformations:

Raw ratios are noisy and non-monotonic: Financial data are messy, so raw financial ratios show irregular, unstable relationships with default risk.

Logit models need monotonic signals: Logistic models perform best when predictors move consistently with PD; noisy ratios cause unstable coefficients and weaker performance.

Nonparametric transforms creates density estimates: RiskCalc's [reference - *ACPMIP*] quantile-based transforms smooth each ratio into a monotonic, PD-consistent scale, improving stability and accuracy.

NON-PARAMETRIC TRANSFORMATIONS

1	fit_nonparametric_transform()	- Learn ratio–default relationships. - Bins a raw ratio into quantiles, computes empirical default rates. - Creates a smooth, density-estimate-based mapping that turns a raw ratio into a score that reflects how its values relate to default risk.
2	apply_nonparametric_transform()	- Apply the learned mapping. - Assigns new ratio values to their quantile-based PD estimates.
3	fit_all_transforms()	- Fit transforms for all ratios. - Runs the above fitting procedure for every raw ratio and stores all transformation curves.
4	transform_train()	- Transform and impute training data. - Applies the transforms to the training set and fills missing transformed values using training-set means.
5	transform_test()	- Transform test data consistently. - Applies the same trained transforms to test/validation data, imputing missing values with training means to avoid leakage.

FINAL MODEL-READY DATASET

Once we finish cleaning the raw data, calculating the financial ratios, checking correlation, and applying the nonparametric transformations, we put together the final set of features for model training.

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```
# Columns to drop due to high correlation
cols_to_drop = [
    "debt_to_equity_raw",
    "gross_margin_raw",
    "equity_ratio_raw",
    "equity_ratio_raw_missing",
    "gross_margin_raw_missing",
    "equity_ratio_raw_missing"
]

df = df.drop(columns=cols_to_drop, errors="ignore")
```

```
# Final raw ratios
final_raw_ratios = [
    "current_ratio_raw",
    'cash_ratio_raw',
    "debt_ratio_raw",
    "operating_margin_raw",
    "roa_raw",
    "roe_raw",
    "asst_turnover_ratio_raw",
    "asst_coverage_ratio_raw"
]

# Missing flags for those ratios
final_missing_flags = [c + "_missing" for c in final_raw_ratios]
```

Ratios removed due to high correlation, followed by the final set of raw financial ratios and their missing- flags.

MODELLING

This section outlines the full modelling pipeline used to estimate one-year PD.

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LOGISTIC REGRESSION

- We built an L2-regularized logistic regression model.
- We trained the model on non-parametrically transformed financial ratios
- We trained using walk forward method over multiple years for preventing future peeking

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Economic Intuition for the Model and Results

Too much debt becomes riskier

Good profits helps the company

Liquidity decline or shortage can cause problems

Poor cash flow can cause distress during crisis

More assets more the company has cushion

Business Impact of the Model

This model helps the bank by turning each company's financial statements into Probability of Default. It can give warning signs like too much debt, low profits, or low liquidity by which the bank can identify risky borrowers before problems happen. This reduces unexpected losses and makes credit decisions more dependable.

The model also helps in improving pricing and making this a less tiring process. The PD is calibrated helping the bank can price loans and improve profitability. The automated scoring helps in application review.The model is simple, transparent, and financially intuitive, it is easy to explain to management and regulators, making it stable and practical for real-world credit-risk use.

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Full Model Specification

The final Probability of Default (PD) model is built using a hybrid approach that combines non-parametric financial-ratio transformations with an L2-regularized logistic regression classifier. We started from the raw balance sheet and income statement data, we focused on a set of financial ratios representing main structure: leverage (debt-to-equity, debt ratio), profitability (operating margin, ROA), efficiency (asset turnover), liquidity (current ratio), and solvency buffer (asset coverage). Each raw ratio was passed through a non-parametric smoothing transformation (T-function), which bins the ratio into quantiles, computes empirical default rates, and fits a **monotonic smoothed curve** to capture the economic relationship between that ratio and default. These transformations remove outliers, reduce skewness, enforce finance-consistent monotonic behavior, and create smooth, stable predictors. The transformed ratios (T(x)) form the final feature matrix that is being fed into logistic regression.

The model uses **L2-regularized logistic regression** via scikit-learn, optimized using the **LBFGS algorithm**, a quasi-Newton gradient-based optimization method that ensures efficient convergence even with multiple predictors. The logistic regression estimates the probability of default via:PD = 1 / (1 + exp(-(β_0 + β^T T(x)))) where β are the learned coefficients and T(x) are the smoothed, monotonic transformed ratios. Regularization prevents coefficient instability and reduces overfitting. Hyperparameters were tuned using **5-fold cross-validation (K-fold CV)** on the transformed training set. In each fold, the dataset was split into five partitions: the model was trained on four folds and validated on the remaining one, rotating until all folds served as validation once. We tested multiple penalty strengths (C = {0.01, 0.1, 1.0}) and selected the configuration that maximized validation AUC and minimized Brier Score.

Full Model Specification

Walk-Forward Time-Based Validation
Because credit risk is *time dependent*, we performed **walk-forward validation** to test the model's performance on future years that were not included in training. We built separate train/test windows: 2007–08 → 2009, 2007–09 → 2010, 2007–10 → 2011, 2007–11 → 2012, and 2007–12 → 2013. For each window, T-functions were refit only on the training years, then applied to the next year's financials, followed by logistic regression estimation. This design helps the real-world underwriting where the bank must score next year's borrowers using only past information.

This model solves the bank's business problem because it transforms messy financial statements into smooth, meaningful risk signals, captures non-linear economic behavior, generates stable probability estimates, and aligns with regulatory expectations for transparency and monotonicity. By combining T-functions, regularization, k-fold CV, walk-forward testing, and an interpretable logistic form, the model produces high-quality PD estimates that directly improve underwriting, pricing, and early warning detection.

EVALUATION

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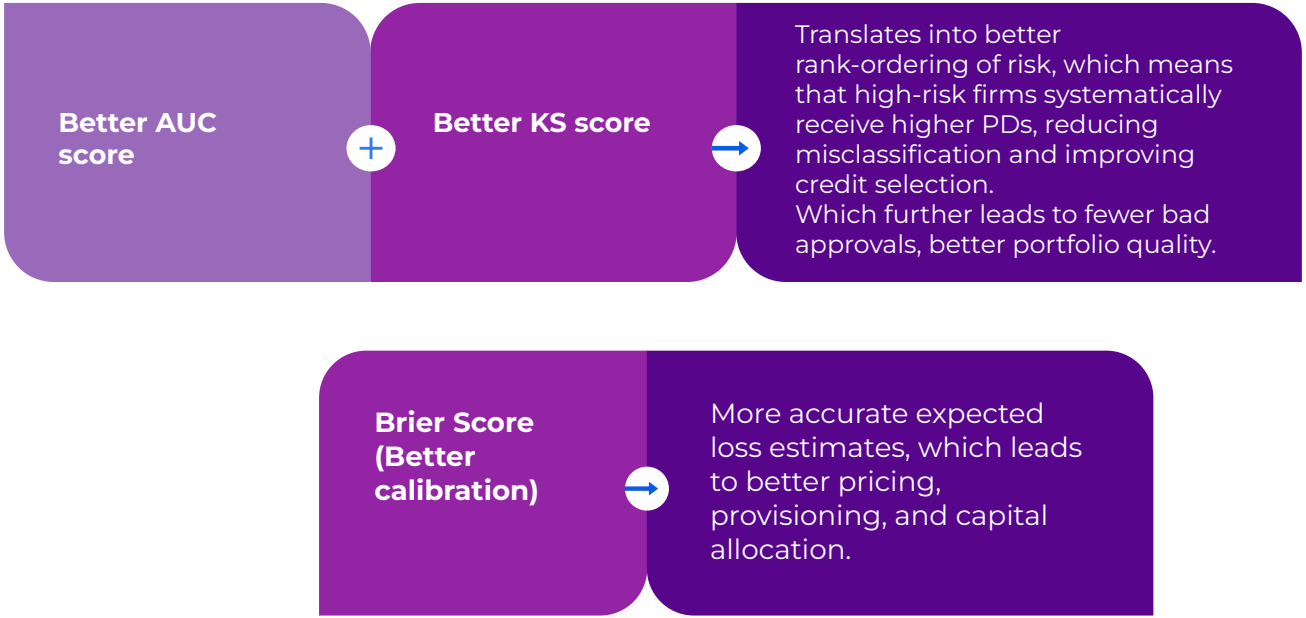
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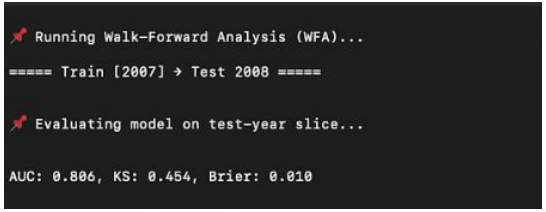
In **evaluation**, the function evaluate_test_year uses three metrics:

ROC AUC	KS Statistic	Brier Score
Measures how well the model ranks risky firms ahead of safe ones.	A credit-risk measure for score separation between good and bad borrowers.	Evaluates how accurate the predicted PDs are as probabilities (calibration).

- These metrics are calculated for every future test year sets in the walk-forward analysis.
- They give us discrimination (ranking), calibration, and robustness.
- Based on the walk-forward performance, we select **C = 0.1 (regularization strength)** as the production hyperparameter.
- These metrics form the complete evaluation framework implemented in our code.

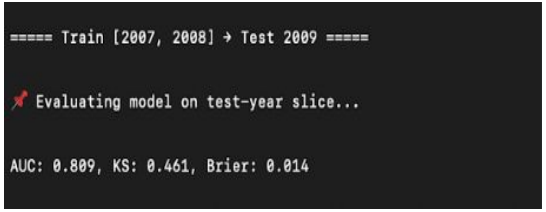
Interpretation of Metric Scores





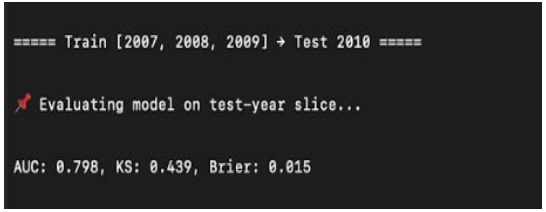
Train 2007 → Test 2008

- AUC 0.806: good ranking ability.
- KS 0.454: indicates solid separation of good vs bad firms.
- Brier 0.010: well-calibrated probabilities.



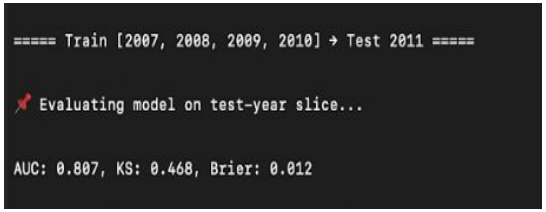
Train 2007-08 → Test 2009

- AUC 0.809: Stable performance.
- KS 0.461: Slightly better separation
- Brier 0.014: still shows good calibration.



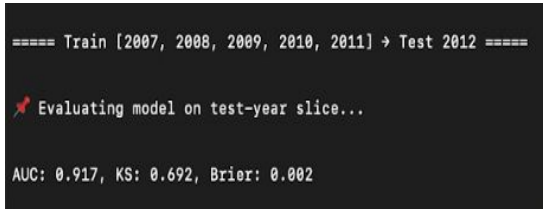
Train 2007-09 → Test 2010

- AUC 0.798: Mild decrease.
- KS 0.439: Slightly better separation
- Brier 0.015: still shows good calibration.



Train 2007-10 → Test 2011

- AUC 0.807: Improves again.
- KS 0.468: Stronger separation.
- Brier 0.012: Better calibration.



Train 2007-11 → Test 2012

- AUC 0.917: Best performance.
- KS 0.692: Very strong score separation.
- Brier 0.002: Excellent calibration.

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DEPLOYMENT

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DEPLOYMENT

How the model will be deployed:

- Integrated into the bank's credit-risk scoring workflow
- **Input**: latest firm financial statements (CSV)
- Automated pipeline:
  - Data cleaning & ratio construction
  - Nonparametric transformation
  - Logistic regression scoring
- **Output**: raw Probability of Default (PD) for each firm

⚠️ **Calibration Required Before Deployment**: Model currently produces uncalibrated PDs. Bank must perform PD calibration to ensure PDs reflect true observed default rates.

Deployment Risks:

Model Drift: The model may become inaccurate over time as economic conditions change, requiring ongoing monitoring and recalibration.

Poor Data Quality: Incorrect or missing financial data can distort PD estimates, so automated data validation is essential before scoring.

Overreliance on the Model: Users may depend too heavily on the PD score, so analyst judgment and override mechanisms must remain part of the decision process.

REFERENCES

- Jeffrey R. Bohn, Roger M. Stein (2009) *Active Credit Portfolio Management in Practice* - Theoretical understanding, model selection and understanding PD modeling principles.
- Provided Cheat Sheets on Brightspace - Deciding on the right ratios, calculating financial ratio and feature engineering steps.
- Lecture slides and class notes - Modeling concepts, evaluation metrics.
- AI assisted tools/Google - Phrasing content and ideas, troubleshoot issues, getting clarity on doubts.

THANK YOU!

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